

Prathmesh Aphale

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SUMMARY

Analytical and detail-oriented Mechanical Engineer with Honors in Robotics and a postgraduate specialization in Data Science with AIML. Skilled in Power BI, SQL, and Python for designing data models, building dashboards, and performing analysis. Experienced in developing insightful reports and visualizations that drive actionable outcomes. Adept at collaborating with cross-functional teams to translate complex data into clear, impactful narratives that enhance operational efficiency and decision-making.

EXPERIENCE

- Kalyani Enterprises, Intern** 12/2023 – 01/2024 | Bhosari, Pune
- Collaborated with production and design teams.
 - Assisted in validating scrubber tank assembly of human cremation system for performance evaluation.
 - Applied analytical problem-solving to ensure project accuracy and timely completion.

EDUCATION

- Edu Plus Now Associated With VIT,** 2025 | Pune
Post graduate program in Data Science with AIML
- Pune Vidyarthi Griha's College Of Engineering And Technology,** 2025 | Pune
Bachelor of Engineering in Mechanical Engineering with Honors in Robotics
| 7.26
- Vishwakarma Vidyalaya 86, HSC** 2021 | Pune
- M.M.E.M.S, SSC** 2019 | Pune
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SKILLS

Programming & Analytics — Python (Pandas, NumPy, Scikit-learn), SQL, **Deep Learning and Neural Networks** — Tensorflow, Keras, NLP, Computer Vision, Time-series modeling, CNN, RNN, LSTM, Pytorch, **Machine Learning** — Regression, Classification, Clustering, **Gen AI** — LLM, Prompt-Engineering, RAG, **Cloud and Data Engineering** — GCP, Azure, AWS, MongoDB, Kafka, PySpark, Databricks, **Data Visualization & BI** — Power BI, Advanced Excel, Matplotlib, Seaborn, **Data Handling** — Data preprocessing, Transformation, Validation, **Tools & Platforms** — Jupyter Notebook, Git/GitHub, Microsoft Office Suite, **Soft Skills** — Analytical Thinking, Data Interpretation, Communication, Collaboration, Attention to detail

PROJECT

Customer Sentiment Analysis on Amazon Reviews

11/2025 – 11/2025

- Built an end to end sentiment analysis pipeline on Amazon reviews with full data cleaning, NLP preprocessing, and EDA.
- Implemented tokenization, stopwords removal, lemmatization, and TF IDF vectorization for clean feature extraction.
- Trained and evaluated a Logistic Regression model using accuracy, precision, recall, and F1 score.
- Created clear visual insights and a polished notebook showcasing strong NLP and ML fundamentals.

Crime Analytics and Prediction

09/2025 – 10/2025

- Built an end-to-end data pipeline from raw Cleveland crime data to a multilabel ML model.
- Designed a MySQL database as a single source of truth and connected it to Power BI for dynamic visualization of crime hotspots and trends.
- Addressed class imbalance by grouping rare crime categories, improving F1-score and fairness.
- Tools: MySQL, Power BI, Python (Pandas, NumPy, Scikit-learn), Matplotlib, Seaborn, Plotly.

Streaming and Watch-time Engagement Analysis.

10/2025 – 10/2025

- Conducted EDA on 5,000 records to uncover patterns in user engagement, content preference, and viewing behavior.
- Analyzed platform, device, genre, and location relationships using visualizations like bar, scatter, and heatmaps.
- Identified trends in binge vs. short-form viewing and highlighted platform performance stability.
- Tools: Python (Pandas, Matplotlib, Seaborn), Jupyter Notebook.

Development of Robotic System for Inspection of Double Bottom Tank,

08/2024 – 06/2025

collaboration with Tolani Maritime Institute

- Designed and simulated a serpentine robotic system for ship tank inspection at Tolani Maritime Institute.
- Contributed to research publication on autonomous inspection and fault detection.
- Implemented motion simulations and feasibility studies using CAD.

Validation of Scrubber Tank Assembly of Human Cremation System, Kalyani Enterprises

12/2023 – 01/2024

- Performed engineering calculations and structural validation to ensure safety and design integrity.
- Collaborated with senior engineers to verify design through tolerance checks and compliance analysis.

CERTIFICATIONS

Computer Integrated Manufacturing (Elite Silver)
NPTEL

Entrepreneurship and IP Strategy (Elite Silver)
NPTEL

Lean Six Sigma: Green Belt Certification

Lean Six Sigma: Yellow Belt Certification